

Week 0 Task 1 Gender Cleaning Report

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Overview

For Week 0 Task 1, the focus was on cleaning the **Gender** column in the Mercer General Emergency Department triage dataset. The task required me to inspect a messy categorical column, identify inconsistent entries, standardize the values, validate the final result, and save a cleaned dataset for later Week 0 tasks. This report documents the full cleaning process, including the original issue, the coding logic, the final human-readable output, and the reasoning behind the changes.

Cleaning Objective

The main objective was to convert inconsistent gender entries into one clear, standardized column. The original dataset contained multiple versions of the same categories, including numeric values and text-based labels. If left uncleaned, these inconsistencies could affect later analysis, visualisation, or modelling. The final goal was to ensure that the **Gender** column was easy to interpret and suitable for clinical data review.

Initial Column Inspection

Before changing the data, I inspected the original **Gender** column to understand the values that were present. This was done using basic pandas commands such as **unique()** and **value_counts()**. This step was important because cleaning decisions should be based on what is actually in the dataset, rather than assumptions.

```
print(df_raw['Gender'].unique())
print(df_raw['Gender'].value_counts(dropna=False))
```

The inspection showed that the column included different representations of male and female patients, such as **0**, **1**, **Male**, **MALE**, **Female**, and **FEMALE**. These values represented the same clinical categories but were recorded inconsistently.

Cleaning Method

I used a function-based cleaning approach. This allowed the notebook to handle several possible data quality issues, including inconsistent capitalization, blank entries, missing values, and unexpected labels. The function first checked whether a value was missing, then converted the value to a lowercase string so that entries such as **Male**, **MALE**, and **male** could be treated consistently.

```
def clean_gender(value):
    if pd.isna(value):
        return -1

    value = str(value).strip().lower()

    if value in ["1", "m", "male", "man", "boy"]:
        return 1
    if value in ["0", "f", "female", "woman", "girl"]:
        return 0
    if value in ["non-binary", "non binary", "nonbinary", "nb", "other"]:
        return 2
    if value in ["unknown", "unspecified", "prefer not to say", "declined"]:
        return -1

    return -1
```

The function created a temporary clean column first, which allowed me to verify the cleaned values before replacing the original column. This reduced the risk of accidentally losing information during the cleaning process.

Original Coding Scheme and Final Readability Fix

The initial cleaning function used numeric codes because numeric categories can be useful for later analysis or machine learning. However, based on feedback, the final version of the cleaned dataset was adjusted so that the **Gender** column used human-readable labels rather than unexplained numeric codes. This makes the dataset easier for mentors, clinicians, and other reviewers to interpret.

Original Code	Meaning	Final Label Used
1	Male	Male
0	Female	Female
2	Non-binary / Other, if present	Would require explicit label or review
-1	Unknown / Unspecified, if present	Would require explicit label or review

```
gender_label_mapping = {
    1: "Male",
    0: "Female"
}

df_raw["Gender"] = df_raw["Gender"].map(gender_label_mapping)
```

This final step ensured that the notebook did not leave the cleaned column as **0** and **1**. Instead, the final output displayed **Male** and **Female**, which improved readability while still preserving the cleaned categories.

Validation After Cleaning

After cleaning, I validated the result using value counts and missing-value checks. This confirmed that the final column contained two readable categories and no missing values in the cleaned output.

```
print(df_raw["Gender"].value_counts(dropna=False))
print(df_raw["Gender"].isnull().sum())
print(df_raw["Gender"].dtype)
```

The updated notebook output showed **Male** and **Female** as the final cleaned values. This confirmed that the earlier numeric codes were successfully converted into readable labels. The cleaned dataset was then saved as a CSV file so it could be reused in later tasks.

Save Output

```
OUTPUT_PATH = '/content/drive/MyDrive/Carisurg Portfolio/Week 0/Data/EmergencyTriageDataset_Gender_Cleaned.csv'
df_raw.to_csv(OUTPUT_PATH, index=False)
```

Saving the cleaned file was important because later Week 0 tasks used the cleaned triage dataset for additional cleaning, visualisation, and clinical interpretation.

Key Cleaning Considerations

- **Inspection before cleaning:** The column was reviewed before transformation so that cleaning decisions were based on the actual values present.
- **Case handling:** Values were converted to lowercase so that different capitalization styles could be cleaned consistently.
- **Missing and unexpected values:** The function included logic for blank, missing, unknown, or unexpected entries instead of silently ignoring them.
- **Readable final output:** The final cleaned column uses **Male** and **Female** rather than numeric codes, making it easier to understand in a clinical data context.
- **Validation:** Value counts and missing-value checks were used after cleaning to confirm that the transformation worked as expected.

Alignment With Feedback

The updated version addresses the main readability issue from the earlier version of the notebook. The original cleaning process successfully standardized the gender values, but the final column was still displayed as numeric codes. The updated version keeps the cleaning logic but converts the final result into readable labels. This means the final dataset is clearer for human review while still being suitable for future encoding if a model requires numeric input later.

Conclusion

The **Gender** column was cleaned by inspecting the original values, standardizing inconsistent entries, handling missing or unexpected values, validating the cleaned results, and saving the updated dataset. The final cleaned column uses human-readable labels, **Male** and **Female**, which improves clarity and makes the dataset easier to interpret. This task reinforced the importance of documenting data cleaning decisions, checking outputs after transformation, and ensuring that cleaned clinical datasets remain understandable to both technical and non-technical reviewers.